Distribute Coins in Binary Tree (medium)

Problem Statement

You are given a `root` node of the binary tree, having `n` nodes, where each node contains the `node.val` number of coins. There are total `n` coins in the whole tree.

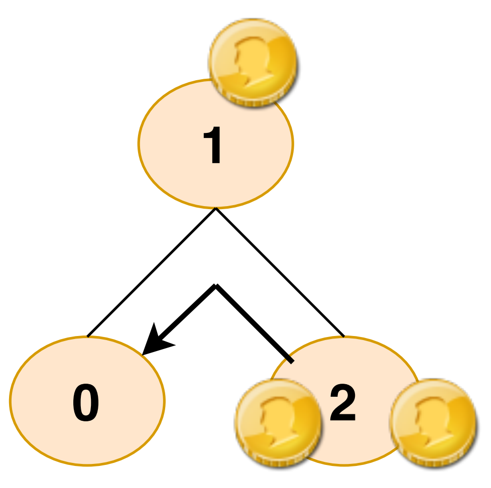
In one move, you can choose any `two adjacent` nodes and move a coin from one node to another. You can either move coins from `parent to child` or `child to parent`.

Return the `minimum` number of `moves` required to ensure every node has `exactly one` coin.

Examples

Example 1:

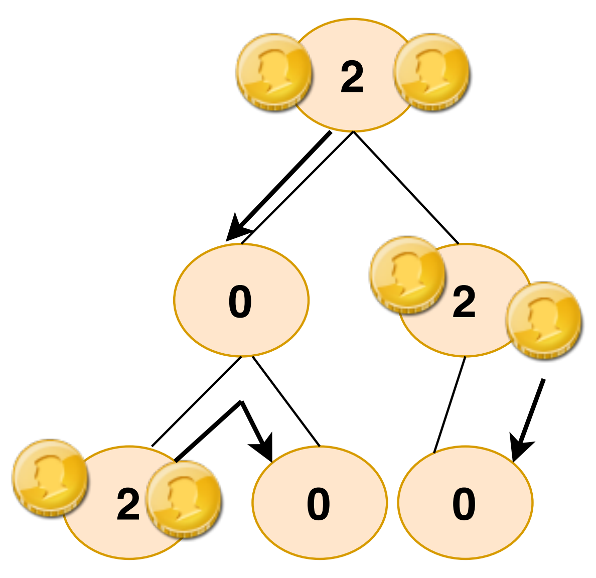
- **Input**: [1,0,2]



- **Expected Output**: 2
- **Justification**: We can move one coin from the right child to the root, and then from the root to the left child. This requires 2 moves to balance the coins.

Example 2:

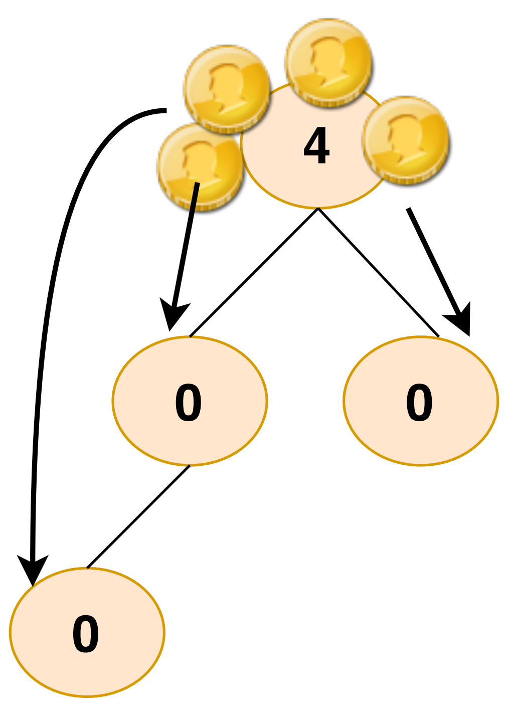
- **Input**: `[2,0,2,2,0,0]`



- **Expected Output**: 4
- **Justification**: Move one coin from the root node to its left node, which requires 1 move. After that, move one coin from root->left->left to root->left->right node, which requires 2 moves. Next, move one coin from root->right to root->right->left, which requires 1 move. Minimum number of moves required is `4`.

Example 3:

- **Input**: `[4,0,0,0]`



- **Expected Output**: 4
- **Justification**: Move two coins from the root to its left child, and then move one coin from this left child to its left child. This requires 3 moves. Move 1 coin to the right child, and this requires 1 move. So, total 4 moves are required.

Try it yourself

Try solving this question here:

Java

```
// class TreeNode {
//     int val;
//     TreeNode left;
//     TreeNode right;

//     TreeNode(int x) { val = x; }

//     TreeNode(int x, TreeNode left, TreeNode right) {
//         this.val = x;
//         this.left = left;
//         this.right = right;
//     }
// }

class Solution {
    public int balanceCoins(TreeNode root) {
        // ToDo: Write Your Code Here.
        return 0; // Return the total number of moves calculated
    }
}
```

Test Case | Run Code Result | Stdout

[1,0,2]

Use Example Testcases | Run | Submit

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